Electroconvulsive therapy

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Subjects

- *Part one*

- ***ECT**: an historical perspective*
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ECT: a historical perspective

Convulsive therapies for major psychiatric illnesses predate the modern therapeutic era, with the use of camphor reported as early as the 16th century and the existence of several accounts of camphor convulsive therapies from the late 1700s to the mid-1800s.

ECT: a historical perspective

Unaware of the history of camphor convulsive therapy, the Hungarian neuropsychiatrist Ladislas von Meduna observed that the brains of people

with epilepsy had more glial cells than other brains.

In contrast, the brains of patients with schizophrenia had fewer glial cells, and von Meduna

hypothesized that there might be a biologic antagonism between convulsions and schizophrenia.

ECT: a historical perspective

In 1934, the first catatonic psychotic patient was successfully treated using intramuscular injections of camphor in oil to produce therapeutic seizures.

Later, other chemicals were used, including pentylenetetrazol.

Insulin coma was also used in the 1940s and 1950s, which also sometimes induced seizures.

ECT: a historical perspective

Lucio Bini and Ugo Cerletti were interested in the use of electricity to induce seizures, and, after a series of animal experiments and observation of the use of electricity commercially, they were able to safely apply current across the heads of animals for this purpose.

In 1938, the first electroconvulsive treatment was administered to a delusional and incoherent patient, who improved with one treatment and remitted after 11 treatments.

ECT: a historical perspective

Electrical induction of convulsive therapy could be made more reliable and shorter-acting than chemically induced convulsive therapies, and, by the early 1940s, it had replaced them.

In 1940, the first use of electroconvulsive therapy (ECT) occurred in the United States.

ECT: a historical perspective

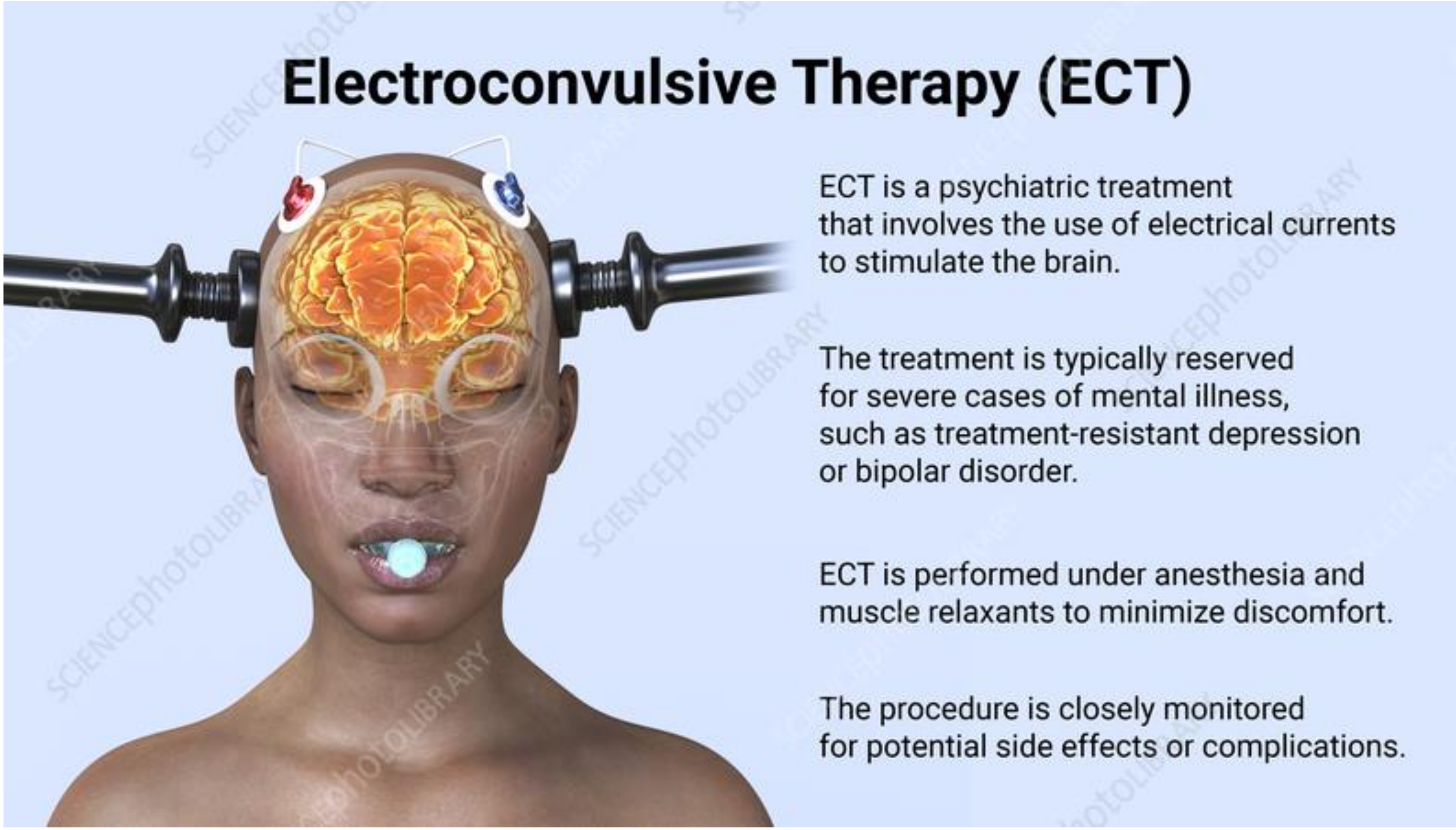
With the widespread use of pharmacologic agents as first-line treatments for major psychiatric disorders, ECT is now more commonly used for patients with resistance to those treatments, except in the case of life threatening illness due to inanition, severe suicidal symptoms, or catatonia.

ECT: a historical perspective

Utilization of ECT has diminished since the middle of the 20th century, but because ECT remains the most effective treatment for major depression and is a rapidly effective treatment for life-threatening psychiatric conditions, ECT, unlike its contemporaneous somatic therapies, such as insulin coma, remains in the active treatment of modern therapeutics.

ECT: a historical perspective

- Its use has shifted from public to private institutions, and it is estimated that approximately 100,000 patients have received ECT annually over the past few decades in the United States .


Electroconvulsive Therapy (ECT)
ECT is a psychiatric treatment that involves the use of electrical currents to stimulate the brain.
The treatment is typically reserved for severe cases of mental illness, such as treatment-resistant depression or bipolar disorder.
ECT is performed under anesthesia and muscle relaxants to minimize discomfort.
The procedure is closely monitored for potential side effects or complications.

Mechanism of Action

The induction of a bilateral generalized seizure is necessary for both the beneficial and adverse effects of ECT.

Although a seizure superficially seems as though it is an all-or-none event, some data indicate that not all generalized seizures involve all the neurons in deep brain structures (e.g., the basal ganglia and the thalamus); recruitment of these deep neurons may be necessary for full therapeutic benefit.

Mechanism of Action

Positron emission tomography (PET) studies of both cerebral blood flow and glucose use have shown that during seizures, *cerebral blood flow, use of glucose and oxygen, and permeability of the blood–brain barrier increase.*

After the seizure, blood flow and glucose metabolism are decreased, perhaps most markedly in the frontal lobes.

Some research indicates that the degree of decrease in cerebral metabolism is correlated with therapeutic response.

Mechanism of Action

ECT affects the cellular mechanisms of memory and mood regulation and raises the seizure threshold .

The latter effect may be blocked by the opioid antagonist naloxone .

Neurochemical research into the mechanisms of action of ECT has focused on changes in neurotransmitter receptors and , recently, changes in second messenger systems .

Mechanism of Action

Virtually every neurotransmitter system is affected by ECT , but a series of ECT sessions results in the downregulation of postsynaptic β-adrenergic receptors , the same receptor change observed with virtually all antidepressant treatments .


ECT

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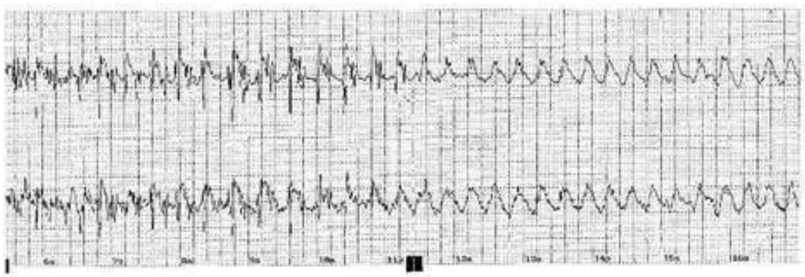
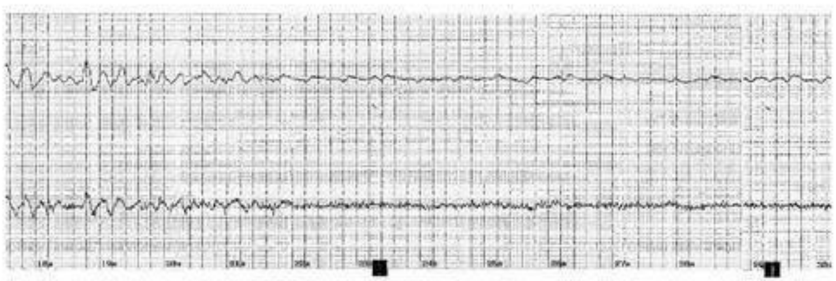
psych scene

EEG reading 2 post ECT





Thymatron System IV S/N: 42133

% Energy Set	45%
Charge Delivered	178.4 mC
Current	0.91 A
Stimulus Duration	2.0 Sec
Frequency	50 Hz
Pulse Width	1.0 mSec
Static Impedance	910 Ohm
Dynamic Impedance	180 Ohm
EEG Endpoint	22 Sec
Average Seizure Energy Index	1887.0 µV2
Postictal Supression Index	52.9%
Maximum Sustained Power	3424.4 µV2
Time to Peak Power	17 Sec

The seizure duration and quality is poor. Note the EEG end point is 22 sec (less than the recommended 25 seconds and a post ictal suppression index of 52.9%. The morphology of the waves compared to the previous reading is poor. A good seizure has a short recruitment phase followed by a spike and wave pattern followed by high amplitude slow waves.

Table 6.7 Psychiatric drugs and ECT

Drug class	Notes	Considerations	Recommendations
Benzodiazepines	May reduce antidepressant efficacy of ECT. Make seizures less likely	Avoid, if possible. Consider non-BDZ hypnotics. Do not suddenly stop if well established	Lowest dose possible. Do not give immediately before ECT (rule of thumb at least 3hrs pre-ECT for last dose)
Antidepressants	May augment antidepressant effect of ECT. Reported prolongation of seizures and tardive seizures	Use low initial stimulus (dose titration methods). Problems reported more for SSRIs	Do not suddenly stop—safely reduce to minimum dose. Prophylaxis may require addition of antidepressant towards end of ECT course (inform ECT team)
Lithium	Reduces seizure threshold. May prolong seizures. May increase post-ictal confusion (case reports only)	Use low initial stimulus (dose titration methods)	Not contraindicated during ECT
Anticonvulsants	Raise seizure threshold	Clarify if drug is for treatment of epilepsy or as a mood stabilizer. Higher doses of ECT may be necessary	If prescribed for epilepsy, continue—do not stop. If a mood stabilizer, continue initially and only reduce if seizure induction is problematic
Antipsychotics	All tend to reduce seizure threshold. Concerns about clozapine (case reports of prolonged and tardive seizures) may be overstated	Use low initial stimulus (dose titration methods)	Clozapine should be withheld 12hrs before any anaesthetic and restarted once fully recovered

Note: ensure the anaesthetist is fully informed of all medications the patient is currently taking.

INDICATIONS

- *Major diagnostic indications:*

- Major depressive disorder

- Mania, including mixed episodes

- Schizophrenia with acute exacerbation or catatonic subtype

- Schizoaffective disorder

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Primary use

- Rapid definitive response required on medical or psychiatric grounds
- Risks of alternative treatments outweigh benefits
- Past history of poor response to medications or good response to ECT
- Patient preference

Secondary use

- Failure to respond to pharmacotherapy in the current episode
- Intolerance of pharmacotherapy in the current episode
- Rapid definitive response necessitated by deterioration of the patient’s condition

Other diagnostic indications

- Parkinson disease
- Neuroleptic malignant disorder

Indications: depressive illnesses

ECT is a proven effective treatment for depression and should not be considered

a last resort option.

It can also be safely administered to patients with physical illnesses, the elderly, and during pregnancy.

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ECT in depression is recommended as a first-line treatment for:

1. the emergency treatment of depression where a rapid definitive response is needed
2. patients with high suicidal risk
3. patients with severe psychomotor retardation and associated problems of eating and drinking or physical deterioration

ECT in depression is recommended as a first-line treatment for:

4. patients who suffer from treatment-resistant depression and who have responded to ECT in previous episodes of illness

5. patients who are pregnant, if there is concern about the teratogenic effects of antidepressants and antipsychotics

6. patients for whom it is the preferred choice of treatment and for whom there are strong clinical indications for its use.

ECT is second-line treatment for:

1. patients with treatment-resistant depression
2. patients who experience severe side-effects from medication, limiting effective treatment
3. patients whose medical or psychiatric condition, in spite of adequate pharmacotherapy, has deteriorated to an extent that raises concern.

ECT is second-line treatment for:

- Old-age psychiatrists may recommend its use in the depressed elderly who have not responded to drug treatments or have suffered unpleasant side-effects.
- Remission rates in clinical trials are 60–70 per cent.

Indications

- *Manic illness*

ECT may be considered for severe mania associated with life-threatening physical exhaustion or treatment resistance.

Indications include the need for a speedy therapeutic response, as a safe alternative to high-dose medications, or if patients have drug-resistant or ‘rapid cycling’ mania.

Indications

Schizophrenia

There are very few indications for ECT in schizophrenia.

Very rarely, it may be used when psychotic symptoms are associated with abnormal motor activity such as catatonic excitement or immobility.

It may be considered if the patient cannot tolerate medications or has failed to respond to adequate doses of antipsychotics including clozapine.

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Contraindications

Although there are no absolute contraindications to ECT , coexisting medical illnesses must be treated.

Close liaison between the psychiatrist, anesthetist , and physicians is necessary .

Contraindications

Pregnancy and old age are not contraindications to ECT, but any contraindications to general anesthesia will apply;

'high-risk' cases with recent myocardial infarction, cerebrovascular accident, or raised intracranial pressure should be assessed on an individual basis in consultation with the anesthetist.

Contraindications

Those with known cardiac disease will need assessment by a physician or cardiologist prior to ECT.

ECG monitoring is necessary during ECT administration, and staff should be adequately trained in cardiopulmonary resuscitation and management of arrhythmias.

Following myocardial infarction or stroke, ECT should be delayed as long as possible and is probably safer after 3 months.

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Contraindications

However, even if there is short-term benefit, after this treatment one must rely on more conventional approaches .

Catatonia

ECT may be considered as a first-line treatment in life-threatening malignant catatonia when treatment with lorazepam has been ineffective .

References

- Kaplan and Sadocks Synopsis of Psychiatry 12th edition
- The Mausley hand book of Psychiatry sixth edition
- Oxford medical hand book of Psychiatry 4th edition